Lecture Notes

Basic Algebraic Structures

Appendix - *Integers*

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CHAPTER

ONE

APPENDIX I - INTEGERS

1.1 Divisibility

Lemma 1.1 (Least Positive Integer) .
The integer 1 is the least positive integer.

Theorem 1.1 (The Well Ordering Principle) .
Every nonempty set S of positive integers contains a least element. That is, there is an element $m \in S$ such that $m \leq x$ for all $x \in S$.

Proof. Let S be a nonempty set of positive integers. If $1 \in S$, then $1 \leq x$ for all $x \in S$, by Lemma 1.1.

Consider now the case where $1 \notin S$, and let L be the set of all positive integers p such that $p < x$ for all $x \in S$. That is,

$$L = \{p \in \mathbb{Z}^+ \mid p < x,\ \forall x \in S\}$$

Since $1 \notin S$, Lemma 1.1 assures us that $1 \in L$. We shall show that there is a positive integer p_0 such that $p_0 \in L$ and $p_0 + 1 \notin L$. Suppose this is not the case. Then we have that $p \in L$ implies $p + 1 \in L$ (since $\neg(a \wedge \neg b) \iff (a \implies b)$). Now, since $1 \in L$, and $p \in L$ implies $p + 1 \in L$, then by Induction Postulate $L = \mathbb{Z}^+$. But this contradicts the fact that S is nonempty (note that $L \cap S =$, by definition of L). Therefore there is a p_0 such that $p_0 \in L$ and $p_0 + 1 \notin L$. Now, we must show that $p_0 + 1 \in S$. We have $p_0 < x$ for all $x \in S$, hence, $p_0 + 1 \leq x$ for all $x \in S$. If $p_0 + 1 < x$ were always true, then $p_0 + 1$ would be in L. Hence, $p_0 + 1 = x$ for some $x \in S$ and $m = p_0 + 1$ is the required least element in S. ■

Def 1.1 (Divisor) .
Let a and b be integers. We say that a **divides** b if there is an integer c such that $b = ac$. If a divides b, we write $a \mid b$.

Theorem 1.2 .
The only divisors of 1 are 1 and -1.

Proof. Suppose a is a divisor of 1. Then $1 = ac$ for some integer c. The equation $1 = ac$ requires $a \neq 0$, so either $a \in \mathbb{Z}^+$ or $-a \in \mathbb{Z}^-$. Consider first the case where $a \in \mathbb{Z}^+$. Then this requires $c \in \mathbb{Z}^+$. So we have $1 \leq a$ and $1 \leq c$, by Lemma 1.1. Then,

$$\begin{aligned} 1 < a &\implies 1 \cdot c < a \cdot c \\ &\implies c < 1 \qquad\qquad \text{(since } ac = 1\text{)}. \end{aligned}$$

But this contradicts with that $1 \leq c$. Thus, $1 = a$ is the only possibility when $a \in \mathbb{Z}^+$.

Similar arguments can be done for $-a \in Z^-$.

Therefore, combining the cases where $a \in \mathbb{Z}^+$ and where $-a \in \mathbb{Z}^-$, we have shown that either $a = 1$ or $a = -1$ if a is a divisor of 1. ■

Theorem 1.3 (The Division Algorithm) .
Let a and b be integers with $b > 0$. Then there exist unique integers q and r such that

$$a = bq + r \text{ with } 0 \leq r < b.$$

Proof. (Existence). Let S be the set of all integers x that can be written in the form $x = a - bn$ for $n \in \mathbb{Z}$, and let S' denote the set of all nonnegative integers in S. Then the set S' is nonempty. If $0 \in S'$, we have $a - bq = 0$ for some q, and hence $a = bq + 0$. If $0 \notin S'$, then by Well Ordering Principle (1.1) S' contains a least element $r = a - bq$, and hence $a = bq + r$, where $r > 0$. Now,

$$\begin{aligned} r - b &= a - bq - b \\ &= a - b(q + 1), \end{aligned}$$

and hence $r - b \in S$. Since r is the least element in S' and $r - b < r$, then $r - b$ must be negative. That is, $r - b < 0$, hence $r < b$. Therefore, there exist such q and r.

(Uniqueness). Suppose that $a = bq_1 + r_1$ and that $a = bq_2 + r_2$, where $0 \leq r_1 < b$ and $0 \leq r_1 < b$. Without loss of generality, we may assume that $r_1 \leq r_2$. This means, $0 \leq r_2 - r_1 \leq r_2 < b$. However, we also have

$$\begin{aligned} 0 \leq r_2 - r_1 &= (a - bq_2) - (a - bq_1) \\ &= b(q_1 - q_2). \end{aligned}$$

That is, $r_2 - r_1$ is nonnegative multiple of b that is less than b. Thus, $r_2 - r_1 = 0$, and hence $r_1 = r_2$. Also, it follow that $bq_1 = bq_2$, where $b > 0$. This implies that $q_1 = q_2$. Therefore, we have shown that $r_1 = r_2$ and that $q_1 = q_2$, and this shows that r and q are unique. Therefore the Theorem. ■

Theorem 1.4 (The Strengthened Division Algorithm) .
Let a and b be integers with $b > 0$. Then there exist

unique integers q and r such that

$$a = bq + r \text{ with } 0 \leq r < |b|.$$

Proof. Left as an exercise. ■

1.2 Prime Factors

Def 1.2 (Greatest Common Divisor (GCD)) .
An integer d is a **greatest common divisor** of a and b if the following conditions are hold:

1. $d \in \mathbb{Z}^+$
2. $d \mid a$ and $d \mid b$
3. $(c \mid a) \wedge (c \mid b) \implies c \mid d$.

Theorem 1.5 .
Let a and b be integers, at least one of them is not 0. Then there exists a unique greatest common divisor d of a and b. Moreover, d can be written as a linear combination of a and b, that is,

$$d = am + bn$$

for $m, n \in \mathbb{Z}$, and d is the smallest positive integer that can be written in this form.

Proof. (Existence). Let a and b be integers, at least one of them is not 0. If $b = 0$, then $a \neq 0$, and hence $|a| > 0$. It is obvious that $d = |a|$ is a gcd of a and b in this case, and either $d = a \cdot (1) + b \cdot (0)$ or $d = a \cdot (-1) + b \cdot (0)$.

Suppose now that $b \neq 0$. Consider the set S of all integers that can be written in the form $ax + by$ for some integers x and y, and let S^+ be the set of all positive integers in S. The set S contains $b = a \cdot (0) + b \cdot (1)$ and $-b = a \cdot (0) + b \cdot (-1)$, hence S^+ is not empty. Then by the Well Ordering Principle 1.1, S^+ has a least element d,

$$d = am + bn.$$

We have d positive, and we shall show that d is a gcd of a and b. By the division algorithm 1.3, there are integers q and r such that

$$a = dq + r \text{ with } 0 \leq r < d.$$

Then,

$$\begin{aligned} r &= a - dq \\ &= a - (am + bn)q \\ &= a(1 - mq) + b(-nq). \end{aligned}$$

Thus, r is in S, and $0 \leq r < d$. By choice of d as the least element in S^+, $r = 0$, and hence $d \mid a$. Similarly, it can be shown that $d \mid b$. Now, suppose that $c \mid a$ and $c \mid b$. Then $a = ch$ and $b = ck$, for some integers h and k. Hence,

$$\begin{aligned} d &= am + bn \\ &= chm + ckn \\ &= c(hm + kn), \end{aligned}$$

which shows that $c \mid d$. Hence, by definition 1.2, $d = am + bn$ is the greatest common divisor of a and b. And from the choice of d as least element of S^+ that d is the smallest positive integer that can be written in this form.

(Uniqueness). Left as an exercise. ■

Remark .
Note that we first choose an element from the set $S = \{ax + by\}$, then we show that it is a gcd.

Theorem 1.5 establishes the existence of $gcd(a, b)$, but does not provide a satisfactory method for finding $gcd(a, b)$. A procedure known as the **Euclidean Algorithm** furnishes a systematic method for finding $gcd(a, b)$ where $b > 0$. It can also be used to find integers m and n such that $gcd(a, b) = am + bn$.

Theorem 1.6 (The Euclidean Algorithm) .
Let a and b be nonnegative integers. Then $gcd(a, b) = r_n$ is the last nonzero remainder after the following successive process:

$$\begin{aligned} a &= bq_0 + r_1, & 0 \leq r_1 < b \\ b &= r_1 q_1 + r_2, & 0 \leq r_2 < r_1 \\ r_1 &= r_2 q_2 + r_3, & 0 \leq r_3 < r_2 \\ &. \\ &. \\ &. \\ r_k &= r_{k+1} q_{k+1} + r_{k+2} & 0 \leq r_{k+2} < r_{k+1}. \end{aligned}$$

Proof. ? - Why we always end up with 0, but not the other nonnegative integers. ■

Def 1.3 (Coprime) .
Two integers a and b are called **coprime** or **relatively prime** if their greatest common divisor is 1.

Theorem 1.7 .
If a and b are relatively prime and $a \mid bc$, then $a \mid c$.

Proof. Suppose that $gcd(a, b) = 1$ and $a \mid bc$. Since $gcd(a, b) = 1$, there exist integers m and n such that $1 = am + bn$, by Theorem 1.5. And since $a \mid bc$, then by definition there exists an integer q such that $bc = aq$. Now,

$$\begin{aligned} 1 = am + bn &\implies c = amc + bnc \\ &\implies c = amc + aqn \text{ since } bc = aq \\ &\implies c = a(mc + qn) \\ &\implies a \mid c. \end{aligned}$$

Therefore the Theorem. ■

Def 1.4 (Prime Integer) .
An integer p is called **prime integer** if $p > 1$ and the only divisors of p are ± 1 and $\pm p$.

Remark .
Note that the condition $p > 1$ makes p positive and ensures that $p \neq 1$. The exclusion of 1 from the set of primes makes possible the statement of Unique Factorization Theorem.

Lemma 1.2 (Euclid's Lemma) .
If p is a prime and $p \mid ab$, then either $p \mid a$ or $p \mid b$.

Proof. Suppose that p is a prime and $p \mid ab$. If $p \mid a$, then there is nothing to show. Suppose then that $p \nmid a$. This implies that $gcd(p, a) = 1$, since the only positive divisors of p are 1 and p. Then by Theorem 1.7, $p \mid b$. Therefore the Theorem. ■

Corollary 1.7.1 .
If p is a prime and $p \mid (a_1 a_2 \dots a_n)$, then p divides some a_i.

Theorem 1.8 (Unique Factorization Theorem) .
Every positive integer n either is 1 or can be expressed as a product of prime integers, and this factorization is unique, except for the order of the factors.

Proof. (Existence). Let P_n be the statement that either $n = 1$ or n can be expressed as a product of primes. We shall prove that the statement P_n is true for all $n \in Z^+$ by the Second Principle of Finite Induction. P_1 is trivially true. Suppose that P_m is true for all positive integers $m < k$. If k is a prime, then k is a product with one prime factor, and hence P_k is true. Suppose k is not a prime. Then $k = ab$, where neither a or b is 1. Therefore, $1 < a < k$ and $1 < b < k$. By the induction hypothesis, P_a and P_b is true. That is, $a = p_1 p_2 \dots p_r$ and $b = q_1 q_2 \dots q_r$, for primes p_i and q_i. Then,

$$\begin{aligned} k &= ab \\ &= p_1 p_2 \dots p_r \cdot q_1 q_2 \dots q_r, \end{aligned}$$

and k is thereby expressed as a product of primes. thus P_k is true, and therefore P_n is true for all positive integers n. (Uniqueness). Suppose that $n = p_1 p_2 \dots p_t$ and $n = q_1 q_2 \dots q_v$ are factorizations of n as products of prime factors p_i and q_j. Then,

$$p_1 p_2 \dots p_t = q_1 q_2 \dots q_v,$$

So $p_1 \mid (q_1 q_2 \dots q_v)$. By corollary 1.7.1, $p_1 \mid q_j$ for some j, and without loss of generality we may assume $j = 1$. However, p_1 and q_1 are primes, so $p_1 \mid q_1$ implies $q_1 = p_1$. This gives us,

$$p_1 p_2 \dots p_t = p_1 q_2 \dots q_v,$$

and therefore,

$$p_2 \dots p_t = q_2 \dots q_v.$$

By repeating with this process we have finally obtain,

$$p_t = q_t \dots q_v.$$

Since the only positive factors of p_t are 1 and p_t, and since q_j is prime, this means that there must be only one q_j on the right-hand side. Hence $v = t$ and $q_t = p_t$. This completes the proof. ■

Theorem 1.9 (Euclid's Theorem on Primes) .
There are infinitely many primes.

Proof. We shall give the proof by contradiction. Suppose that there are a finite number n of primes, $p_1, p_2, \dots p_n$. Now, consider the integer $m = p_1 p_2 \dots p_n + 1$. It is clear

that the remainder in the division of m by any prime p_i is 1, so each p_i is not a factor of m. Thus, there are two possibilities, either m is itself a prime, or it has a prime factor that is different from every one of the p_i. In either case, we have a prime integer that was not in the list. Therefore, there are more than n primes and this completes the proof. ■

1.3 Congruence of Integers

Def 1.5 .
Let n be a positive integer, $n > 1$. For integers x and y, x is said to be **congruent** to y **modulo** n if $x - y$ is a multiple of n, and we write

$$x \equiv y \pmod{n}.$$

Remark .
Note that $x \equiv y \pmod{n} \iff x - y = nq$ for some $q \in \mathbb{Z}$.

Theorem 1.10 .
The relation of congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Reflexive.

$x \equiv x \pmod{n}$ since $x - x = (n)(0)$.

Symmetric.

$$\begin{aligned} x \equiv y \pmod{n} &\implies x - y = nq \quad \exists q \in \mathbb{Z} \\ &\implies y - x = n(-q), \ -q \in \mathbb{Z} \\ &\implies y \equiv x \pmod{n}. \end{aligned}$$

Transitive.

$$\begin{aligned} x \equiv y \pmod{n} \text{ and } \quad & y \equiv z \pmod{n} \\ &\implies x - y = nq,\ y - z = nk \\ &\exists q, k \in \mathbb{Z} \\ &\implies x - z = x - y + y - z \\ &= n(q + k),,\ q + k \in \mathbb{Z} \\ &x \equiv z \pmod{n}. \end{aligned}$$

Remark .
Remember that any equivalence relation, the equivalence classes for congruence modulo n form a *partition* of $\mathbb{Z}$; that is they separate $\mathbb{Z}$ into mutually disjoint subsets. These subsets are called **congruence classes**. For example, when $n = 4$, the congruence classes appears as:

$$\begin{aligned} [0] &= \{\ldots, \quad -4, \quad 0, \quad 4, \quad \ldots\} \\ [1] &= \{\ldots, \quad -5, \quad 1, \quad 5, \quad \ldots\} \\ [2] &= \{\ldots, \quad -6, \quad 2, \quad 6, \quad \ldots\} \\ [3] &= \{\ldots, \quad -7, \quad 3, \quad 7, \quad \ldots\} \end{aligned}$$

Theorem 1.11 .
If $a \equiv b \pmod{n}$ and x is any integer, then $a + x \equiv b + x \pmod{n}$ and $ax \equiv bx \pmod{n}$.

Proof. Left as an exercise. ■

Remark .
We can use cancellation on addition but not on multiplication. By this I mean,

$$a + x \equiv a + y \pmod{n} \implies x \equiv y \pmod{n},$$

but

$$ax \equiv ay \pmod{n} \not\implies x \equiv y \pmod{n}.$$

The second one can be easily shown by giving a counter-example.

Theorem 1.12 (Cancellation Law) .
If $ax \equiv ay \pmod{n}$ and $gcd(a, n) = 1$,
then $x \equiv y \pmod{n}$.

Proof. Suppose $ax \equiv ay \pmod{n}$ and $gcd(a, n) = 1$. Then,

$$\begin{aligned} ax \equiv ay \pmod{n} &\implies n \mid (ax - ay) \\ &\implies n \mid a(x - y) \\ &\implies n \mid (x - y) \text{ by Theorem 1.7} \\ &\implies x \equiv y \pmod{n}. \end{aligned}$$

This completes the proof. ■

Theorem 1.13 (Substitution Properties) .
Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$.
Then, $a + c \equiv b + d \pmod{n}$ and $ac \equiv bd \pmod{n}$.

Proof. Left as an exercise. ■

Ex. 1.3.1. Find the integer x for $58^{23} \equiv x \pmod{9}$.

Solution. Since $58 \equiv 4 \pmod 9$, then by Theorem 1.13, we have $58^{23} \equiv 4^{23} \pmod 9$. Also since $4^{23} = 4^2 \cdot (4^3)^7$, then

$$\begin{aligned} 58^{23} &\equiv 4^{23} \pmod 9 \\ &\equiv 4^2 \cdot (4^3)^7 \pmod 9 \\ &\equiv (16)(64)^7 \pmod 9 \\ &\equiv (7)(1)^7 \pmod 9 \\ &\equiv 7 \pmod 9. \end{aligned}$$

This solves the problem. □

Theorem 1.14 (Linear Congruences) .
If a and n are co-prime, the congruence $ax \equiv b \pmod n$ has a solution x in the integers, and any two solutions in $\mathbb{Z}$ are congruent to modulo n.

Proof. Since a and n are co-prime, there exist integers s and t such that

$$\begin{aligned} &1 = as + nt \\ &\implies b = asb + ntb \\ &\implies a(sb) - b = n(-tb) \\ &\implies n \mid [a(sb) - b] \\ &\implies a(sb) \equiv b \pmod n. \end{aligned}$$

Thus, $x = sb$ is a solution to $ax \equiv b \pmod n$.

(Uniqueness). To complete the proof, suppose that both x and y are integers that are solutions to $ax \equiv b \pmod n$. Then we have,

$$ax \equiv b \pmod n \quad \text{and} \quad ay \equiv b \pmod n.$$

Then by symmetric and transitive properties of congruence modulo n, we conclude that these relations imply

$$ax \equiv ay \pmod n.$$

Since $gcd(a, n) = 1$, then by Cancellation Law 1.12 we get $x \equiv y \pmod n$. Therefore, any two solutions in $\mathbb{Z}$ are congruent modulo n. ■

Theorem 1.15 .
Let m and n be co-prime and a and b be integers. There exists an integer x that satisfies the system of congruences

$$\begin{aligned} x &\equiv a \pmod m \\ x &\equiv b \pmod n. \end{aligned}$$

Furthermore, any two solutions x and y are congruent modulo mn.

Proof. (Existence). Suppose $gcd(m, n) = 1$. Let x be a solution to the first congruence in the system

$$\begin{aligned} x &\equiv a \pmod m \\ x &\equiv b \pmod n. \end{aligned}$$

Thus, $x = a + mk$ for some integer k, and this must be such that $a + mk \equiv b \pmod n$ or, $mk \equiv b - a \pmod n$. Since $gcd(m, n) = 1$, Theorem 1.14 guarantess the existence of such an integer k, and $x = a + mk$ satisfies the system.

(Uniqueness). Left as an exercise. ■

Theorem 1.16 (Chinese Remainder Theorem) .
Let $n_1, n_2, \ldots n_m$ be pairwise co-prime. There exists an integer x that satisfies the system of congruences

$$\begin{aligned} x &\equiv a_1 \pmod{n_1} \\ x &\equiv a_2 \pmod{n_2} \\ &. \\ &. \\ &. \\ x &\equiv a_m \pmod{n_m}. \end{aligned}$$

Furthermore, any two solutions x and y are congruent modulo $n_1 n_2 \ldots n_m$.

Proof. Left as an exercise. ■

Theorem 1.17 (Fermat's Little Theorem) .
If p is a prime number, then for any integer a we have

$$a^p \equiv a \pmod p.$$

Proof. **RETURN!** ■

1.4 Congruence Classes

We have observed that, in connection with the relation of congruence modulo n, there are n distinct congruence

classes. Let $\mathbb{Z}_n$ denote this set of classes:

$$\mathbb{Z}_n = \{[0], [1], \ldots [n-1]\}.$$

Theorem 1.18 .
Consider the rule given by

$$[a] + [b] = [a + b].$$

Then,

a. This rules defines an addition that is a binary operation on $\mathbb{Z}_n$.

b. Addition is associative in $\mathbb{Z}_n$.

c. Addition is commutative in $\mathbb{Z}_n$.

d. $\mathbb{Z}_n$ has the additive identity $[0]$.

e. Each $[a]$ in $\mathbb{Z}_n$ has $[-a]$ as its additive inverse in $\mathbb{Z}_n$.

Proof.

a. It is clear that the rule $[a]+[b] = [a+b]$ yields and element of $\mathbb{Z}_n$. In other words, closure is obvious, but we need to show that the operation is well-defined. To do this, suppose that $[a] = [x]$ and $[b] = [y]$. Then $[a] = [x] \implies a \equiv x \pmod{n}$ and $[b] = [y] \implies b \equiv y \pmod{n}$. Then by Theorem 1.13, $a+b \equiv x+y \pmod{n}$, and therefore $[a+b] = [x+y]$. Therefore, the rule is well-defined.

b.

$$\begin{aligned}
[a] + ([b] + [c]) &= [a] + [b + c] \\
&= [a + (b + c)] \\
&= [(a + b) + c] \\
&= [a + b] + [c] \\
&= ([a] + [b]) + [c].
\end{aligned}$$

c. Left as an exercise.

d. Since addition is commutative in $\mathbb{Z}_n$ and $[a]+[0] = [a+0] = [a]$.

e. Left as an exercise.

■

Remark .
Note that the key step in part b. of the proof is the fact that addition is associative in $\mathbb{Z}$.

Theorem 1.19 .
Consider the rule for multiplication in $\mathbb{Z}_n$ given by

$$[a][b] = [ab].$$

Then,

a. Multiplication as defined by this rule is a binary operation on $\mathbb{Z}_n$.

b. Multiplication is associative in $\mathbb{Z}_n$.

c. Multiplication is commutative in $\mathbb{Z}_n$.

d. $\mathbb{Z}_n$ has the multiplicative identity $[1]$.

Proof. Left as an exercise. ■

Remark .
A multiplication table for $\mathbb{Z}_4$ is shown below. The third row of the table shows that $[2]$ is a nonzero element of $\mathbb{Z}_4$ that has no multiplicative inverse. Another interesting point is that the equality $[2][2] = [0]$ shows that the product of nonzero factors may be zero.

$\times$	$[0]$	$[1]$	$[2]$	$[3]$
$[0]$	$[0]$	$[0]$	$[0]$	$[0]$
$[1]$	$[0]$	$[1]$	$[2]$	$[3]$
$[2]$	$[0]$	$[2]$	$[0]$	$[2]$
$[3]$	$[0]$	$[3]$	$[2]$	$[1]$

Theorem 1.20 .
An element $[a]$ of $\mathbb{Z}_n$ has a multiplicative inverse in $\mathbb{Z}_n$ if and only if a and n are co-prime.

Proof. ($p \implies q$). Suppose first that $[a]$ has a multiplicative inverse $[b]$ in $\mathbb{Z}_n$. Then $[a][b] = [1]$. This means that $[ab] = 1$, and hence $ab \equiv 1 \pmod{n}$. Thus, $ab - 1 = nq$ for some integer q and $a(b) + n(-q) = 1$. Then by Theorem 1.5, $gcd(a, n) = 1$.

($p \impliedby q$). Conversely, if $gcd(a, n) = 1$, then Theorem 1.14 guarantees the existence of a solution s to the congruence $as \equiv 1 \pmod{n}$. Therefore, $[a][s] = [1]$ and $[a]$ has a multiplicative inverse $[s]$ in $\mathbb{Z}_n$. ■

Corollary 1.20.1 .
Every nonzero element of $\mathbb{Z}_n$ has a multiplicative inverse if and only if n is prime.

1.5 Exercises

Ex. 1.5.1.

a. Prove that $(\forall n \in \mathbb{Z}^+)10^n \equiv 1 \pmod 9$.

b. Prove that a positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Ex. 1.5.2.

a. Prove that $(\forall n \in \mathbb{Z}^+)10^n \equiv (-1)^n \pmod{11}$.

b. Prove that a positive integer z is divisible by 11 if and only if 11 divides $a_0 - a_1 + a_2 - \cdots + (-1)^n a_n$, where $z = a_n 10^n + a_{n-1}10^{n-1} + \cdots + a_1 10 + a_0$.

BIBLIOGRAPHY